Predictive Loyalty Modeling and Geo-Based Logistic Hub Optimization

Md Nazmul Islam Razib
Department of Microdata Analysis
Högskolan Dalarna
Borlänge, Sweden
e-mail: h24mdnra@du.se

Md Ariful Ahsan
Department of Microdata Analysis
Högskolan Dalarna
Borlänge, Sweden
e-mail: h24mdaah@du.se

***Abstract*— Effective logistic planning and customer retention are central challenges faced by e-commerce companies and supply chain systems. This study presents a data-driven framework to identify loyal customers and use their geographic patterns to suggest potential logistic hub locations. Rather than using traditional classification models, we utilized Recency, Frequency and Monetary (RFM) metrics to segment customer loyalty. Then loyal customers were geographically clustered using K-Means clustering, with the optimal 'k' which determined through the Elbow method and Silhouette score analysis. The centroid of these clusters defines as predictive indicators for future logistic hub expansion. This approach integrates loyal customer modeling with geospatial analytics which offers logistic companies' actionable insights for hub expansion planning based on demand concentration and customer value.**

Keywords- Logistics; RFM; Loyal city; Customer; Geographic Analysis; K- Means Clustering; Locations.

I. INTRODUCTION

In today's competitive logistics landscape having an organized logistics system is more important than ever. Companies must not only deliver products quickly and efficiently but also ensure a smooth and satisfying customer experience. A significant challenge in this area is determining the optimal placement of logistics hubs to maximize efficiency and customer reach. The location of these hubs directly impacts delivery times, operational costs and ultimately customer satisfaction and business performance.

At the same time, understanding customer behavior and identifying that loyal customers has become crucial. Loyal customers are often the backbone of a business. They tend to shop more frequently, spend more per transaction, and are more likely to recommend the brand to others. Providing these customers with superior service or faster delivery can significantly enhance customer retention and boost overall sales.

In this report, we propose a data-driven approach that combines customer loyalty analysis with location intelligence to inform logistics decision-making. First, we analyze customer behavior using three essential indicators: purchase frequency, recency of the last purchase, and total amount spent. These metrics help us evaluate each customer's value and level of loyalty. We then employ clustering algorithms to group customers into segments based on these factors, enabling us to focus specifically on high-value, loyal customers.

Once we identify the loyal customer segment, we extract the geographical locations of their orders. The next step is to cluster these order locations based on their coordinates. This spatial clustering enables us to uncover patterns in the concentration of loyal customers, allowing us to recommend potential locations for logistics hubs that align with actual customer demand.

By integrating customer analytics with geographic insights, our approach aims to create a more responsive and efficient logistics network that prioritizes the needs of the most valuable customers. This strategy not only improves delivery performance but also strengthens customer relationships and drives long-term business growth.

II. LITERATURE REVIEW

Nowadays, logistics analytics has increasingly embraced data-driven techniques to improve customer retention and operational efficiency. Central to these efforts are data cleaning and geo-validation, as inaccuracies in geographic information can greatly distort subsequent analyses and predictive modeling. Previous studies highlight the importance of accurate geo-validation to ensure reliable customer segmentation and meaningful regional insights. [2], [7].

One of the most established techniques for customer segmentation is RFM (Recency, Frequency, Monetary) analysis. This method quantifies customer behavior based on how recently they made a purchase, how frequently they place orders, and the monetary value of their purchases. By using RFM we can identify and target high- values, loyal customer segments [3].

Recent research illustrates that integrating RFM analysis with clustering algorithms, such as k-Means, enables the discovery of nuanced customer subgroups. This integration allows for more targeted retention and loyalty strategies [4]. In particular, k-Means clustering has proven to be effective in grouping customers based on their behavioral patterns, leading to practical applications in personalized marketing and logistics planning. [1], [4].

Geographic analysis is also becoming increasingly important in logistics strategy, mainly for optimizing the placement of hubs distribution and planning service expansions. Studies suggest that analyzing the spatial distribution of loyal customers and measuring their proximity to existing logistics hubs can uncover underserved regions and opportunities for hub placement [6], [8]. A practical approach involves using the centroids of clusters formed from loyal customer locations which provides a data driven basis to identify central hub locations that shows high value demand zones.

RFM, clustering and geospatial analysis shows a leading framework in predictive loyalty modeling and logistics network optimization. This research builds on such frameworks by applying RFM, K-Means clustering to a logistics dataset [6]. Validating geographic data and predicting expansion strategies based on clustering aligns with recent trends in logistics research where machine learning shows more effective and intelligent decision making [1], [5].

III. Method Description

A. The Dataset

The dataset we used in this report was sourced from Kaggle which is a popular platform for data science competitions and datasets. This is the source link of the dataset https://www.kaggle.com/datasets/pushpitkamboj/logistics-data-containing-real-world-data The selected datasets refers to logistic or shipping data and included various features related to orders, sales, delivery times, customer locations and product information.

We also looked into the correlation between our whole dataset variables (Figure 1). As we can see the labels has the least correlation.

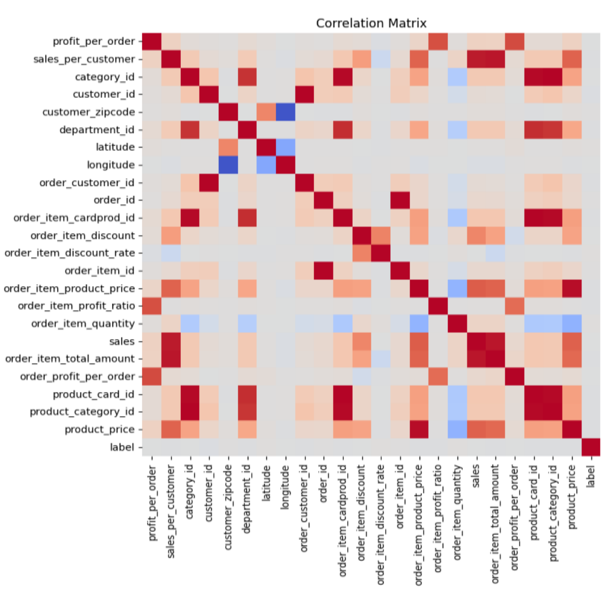


Figure 1: Correlation of All Variables

B. Tools

The project was implemented using the 'Python' programming language. Where we used libraries such as- pandas, NumPy, Scikit-learn, googlemaps, folium, plotly, Matplotlib and seaborn and some other necessary tools.

C. Data Preprocessing

The dataset was first loaded with pandas. Initial exploration was conducted to understand the structure of the dataset and identify missing or inconsistent values. And we found the following:

The dataset is a logistic database containing data of 15,549 observations and 41 variables. However, to find out loyal customers and identify their locations we are only using the following variables:

TABLE I. ESSENTIAL VARIABLE DESCRIPTION

Variable Name	**Description**
customer_id	Customer unique identification
order_country	Destination country of the order
order_city	Destination city of the order
sales	Value in sales
order_date	Date on which the order is made
sales_per_customer	Total sales per customer made per customer
order_item_product_price	Price of products without discount
latitude	Latitude corresponding to location of store
longitude	Longitude corresponding to location of store

Figure 2: Description of the Variables

The dataset has no empty values. We examined the data and didn't find any outliers. There were some mismatches between order city and order country. To treat them we used geo coding and added correct country to the dataset. In this step we focused on detecting and correcting inconsistent or missing country information in the dataset, particularly where the order city and order country fields appeared mismatched. We utilized the Google Maps geocoding API to validate and correct the country data based on each city name. Also, a Google Maps client was created using a valid API key. For each unique city, the correct country was fetched using the previously defined function and stored in a dictionary. A short delay for 0.2 seconds was added between API calls to comply with Google rate limit. Then the order country column was updated using the city to country mapping.

In finding out customer loyalty with RFM, the following variables were very much correlated (Figure 3): sales, sales per customer, and order item product price while calculating the monetary value. We used the sales variable and dropped the other two as sales has the highest correlation.

Sales help to segment high spender that's why we kept sales to calculate the monetary value to find the loyal customers.

Figure 3: Correlation of Sales Related Variables

As the monetary value shows how much a customer spends while calculating RFM. That's why the sales variable was used as the monitory indicator, while the order date was used as recency indicator. We counted orders for each customer and used as frequency indicator. Using these three key factors we modelled our customer's loyalty.

D. Data Mining Method

We used the RFM model to identify our loyal customer from the dataset. The recency (R) is calculated from the order date, the frequency (F) is calculated by grouping the data by customer id and counting their frequency of order and finally the monetary (M) is calculated as sum of the sales value ordered by the customer. The calculated value of recency, frequency and monetary were then used to cluster customers. We used K-Means clustering. As we had only 3 variables which is less than 10, so there was no need for dimension reduction.

Before fitting our data to the model, we scaled them using Standard scaler. To find out the optimal number of clusters, we used Silhouette method. In that cases, 2 looked like a reasonable number for the clustering (Figure 4). As cluster 2 has the highest score over 0.9, that's why we choose k = 2 as our optimal cluster.

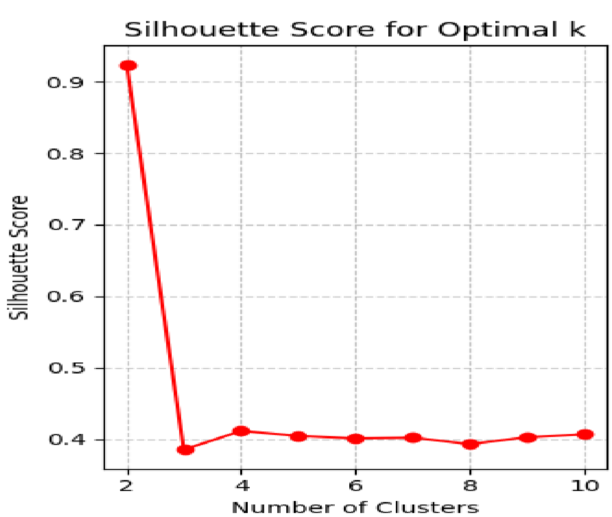


Figure 4: Identifying Optimal Clusters of Customers

For each cluster we further analysed them based on the three key factors: Recency, Frequency and Monetary value (Figure 5).

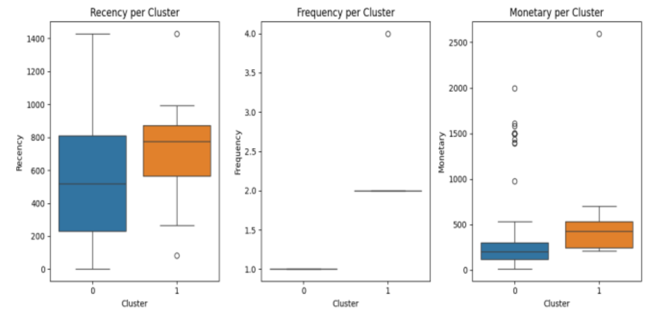


Figure 5: Cluster Distribution

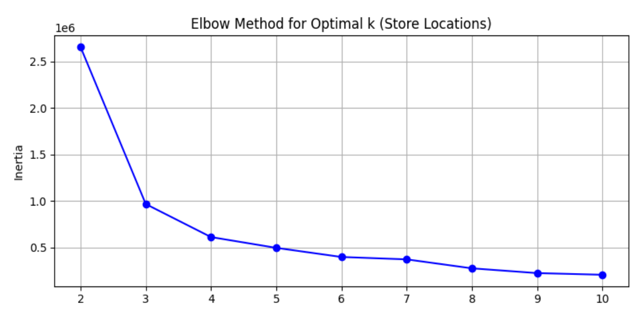
For cluster 0, we have low recency, meaning the customers in this cluster has not recently placed orders. We calculated recency based on order placed date with the most recent order date in our dataset. Customers in cluster 1 has high recency, high frequency and high monetary as well. We can termed this cluster as loyal customers cluster.So, we treated cluster 1 as loyal. The following table shows the customer count in each cluster (Figure 6).

TABLE II. CUSTOMER COUNT IN CLUSTER

Cluster	Customer Count
0	15491
1	28

Figure 6: Customers in Each Cluster

In the next step we further analysed customers from cluster 1. We identified the order location and converted the value into latitude and longitude format using googlemaps library. We also took store locations in considerations, when we plotted the store locations, we found the locations were very highly dense (Figure 8). So, we applied elbow and silhouette score to find the optimal cluster number. And we find that the optimal cluster is 3 for the centroid. The score of silhouette is highest on 3 (Figure 7)



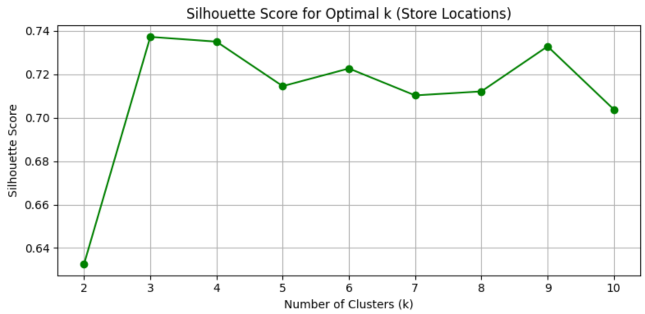


Figure 7: To Find the Optimal K for Centroids of Store Locations.

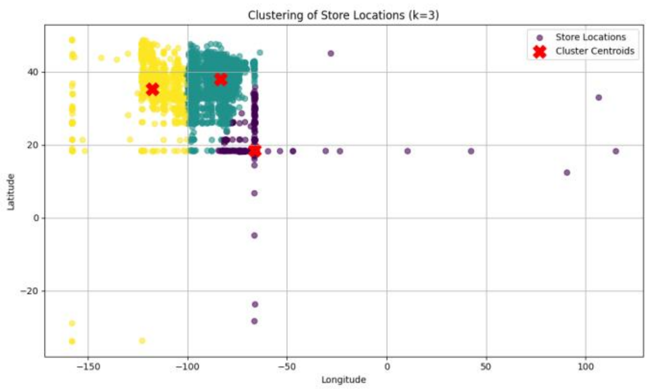


Figure 8: Centroids for Store Locations

For most loyal customers' order city, we applied both elbow and silhouette method to identified the optimal K (cluster number) (Figure 9) which is 3. We used matplotlib, seaborn to plot the clustered loyal cities. We also used plotly to visualize the interactive world map of loyal cities. Then we again applied K-Means clustering based on the loyal cities' geolocation data to identify central hub areas. We grouped the dataset by each cluster from K-Means clustering.

To better understand the nature of each logistic hub cluster, we computed a summary for every cluster. For each group or cluster, it calculatesthe number of customers in each cluster based on loyal customer id, the average latitude and longitude of loyal cities and the most frequent city among loyal cities location in that cluster. And we use folium to plot the clustered loyal cities' geolocation on the map with the centroid of new logistic hub.

IV. Results and Analysis

After analysis, we found 28 most loyal customers from our data set. After taking them as loyal customers from the cluster 1, we went further to understand their geographical distribution, we filtered the original dataset to do further process with only order placed by these loyal customers. From that filtered data we extracted the unique combinations of customer id, order city and order country to show where these loyal customers are receiving shipments. As a result, we found 57 unique city customer combinations. That indicates that several loyal customers have placed orders to multiple locations. Which gives us actionable insight into where the most valuable customers are located.

To identify the optimal logistic hub locations based on the loyal customer behavior, we first performed clustering analysis on their geographical order locations (latitude and longitude). After using Silhouette Score analysis we found a noticeable high point k = 3 (Figure 9) which indicates a good balance between model complexity and explained variance. Also, in Silhouette Score for k values from 2 to 10 (Figure 9) the score peaked at k = 3 which confirms that three clusters provide the best defined and well separated grouping of loyal customer locations. The Silhouette Score for K=3 is: 0.6663.

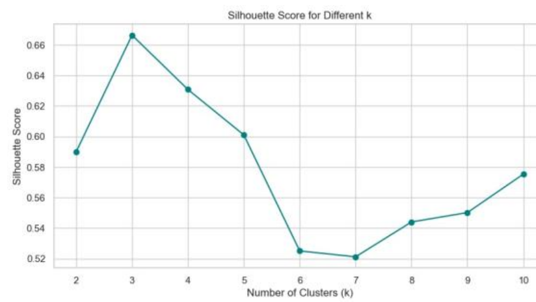


Figure 9: Identifying Optimal Cluster for Loyal City

An interactive world map (Figure 10) shows the cluster along with detailed hover information.

Figure 10: Geographic Clustering of Loyal Customer Using k=3

The cluster summary revealed distinct geographic groupings of loyal customers from each cluster. Each cluster had a central location calculated from the average latitude and longitude and a dominant city which has the most loyal customers in that cluster where the customer ordered from. These cities represent potential candidates for logistics hubs. For example, cluster 0 has 23 size and the majority of loyal customers orders from 'Kars' city and cluster 1 (size – 14) from Aurangabad, then cluster 2 (size – 20) from Managua.(Figure 11)

Figure 11: Predicted New Logistic Hub Points Based on Customer Loyalty Geo- Clustering

V. Conclusion

We used unsupervised learning methods as there was no clear label for the loyal customers in our dataset. Using K-Means clustering we classified customers into 4 clusters, among which we choose 28 customers from the most loyal cluster for further analysis. We identified the order cities and clustered the locations into 3 geographic clusters which represent potential candidates for future logistic hub expansion or regional marketing strategies. This methodology not only highlights the value of data-driven customer segmentation but also shows how unsupervised learning can be leveraged to inform strategic business decisions with proper insights such as- optimizing delivery networks and tailoring regional engagement strategies. The same methodology can be applied for a different dataset to identify loyal customers and suggest optimal hub expansions.

In short, the combination of RFM analysis and clustering provides a strong framework to uncover customer loyalty patterns and support strategic planning for business growth.

References

[1] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed., Morgan Kaufmann, 2011.

[2] M. A. Goodchild and L. Anselin, "Spatial data analysis and GIS: A review of progress and challenges," *Geographical Information Systems*, vol. 1, pp. 239–251, 1992.

[3] P. S. Fader, B. G. S. Hardie, and K. L. Lee, "RFM and CLV: Using iso-value curves for customer base analysis," *Journal of Marketing Research*, vol. 42, no. 4, pp. 415–430, 2005.

[4] S. M. Lee, S. Kang, and J. Park, "Customer segmentation and strategy development based on RFM model and K-means clustering," *Expert Systems with Applications*, vol. 38, no. 3, pp. 15274–15285, 2011.

[5] M. Kumar, S. Sharma, and J. Singh, "Customer churn prediction in logistics using machine learning algorithms," *International Journal of Logistics Management*, vol. 32, no. 2, pp. 123–140, 2021.

[6] A. Gupta and T. Lee, "Optimizing logistics hub locations using K-means clustering," *IEEE Access*, vol. 9, pp. 54321–54332, 2021.

[7] B. Smith and L. Wang, "Data-driven logistics: The role of analytics in supply chain management," *Transportation Research Part E*, vol. 138, pp. 101–115, 2020.

[8] S. D. Kim, "Geospatial analysis for optimal logistics hub placement," *Journal of Transport Geography*, vol. 82, pp. 102557, 2020.

[9] Pushpit Kamboj, "Dataset, Real World Logistic Data," Kaggle, 2022. Available: https://www.kaggle.com/datasets/pushpitkamboj/logistics-data-containing-real-world-data